6/5/2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025		
Maximum Marks: 50 Examination: End-Semester Examination Duration: 2 Hrs 40 min		
Programme code: 06 Programme: B.Tech Computer Engineering	Class:SY	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/Center: COMP	
Course Code: 216U01C403	Name of the Course: Relational Database Management Systems PWD	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt the following (MCQ for 10 marks) I. The ability to modify physical-level schema without affecting the logical-level/view-level schema is called: a) Logical Data Independence b) View Data Independence c) Physical Data Independence d) Application Data Independence II. In the company XYZ, employees are given freedom to choose upto 3 projects to work on. Usually, a team of 4-5 people work on a given project. Suppose worksOn is a relationship type with two participating entity types Employee and Project. What is the appropriate cardinality ratio for Employee:Project a) 1:N b) N:1 c) M:N d) 1:1 III. Given two union compatible relations X(A,B) and Y(C,D). What is the result of the operation? $X \bowtie_{(A=C)\wedge(B=D)} Y$ a) X∪Y b) X∩Y c) X−Y d) X÷Y IV. Consider the following query SELECT rollNo as identifier FROM student UNION SELECT empId as identifier FROM professor Which of the following statements is correct regarding the above query?	10

a) The query executes only if rollNo of student and empId of professor have same data types
b) The query executes in all cases (irrespective of the data types of rollNo and empId)
c) The query does not execute at all
d) None of the above

V. Consider two instances of the relations R1(A, B) and R2 (C, D) with the number of rows 10 and 8, respectively. What is the least possible number of tuples in the result of the following query?

SELECT * FROM R1 FULL OUTER JOIN R2 ON R1.A = R2.C

a) 0
b) 18
c) 8
d) 10

VI. We use _____ to find functional dependencies (FDs) logically implied by the given set of FDs . Choose the most appropriate option:

a) Trivial FDs
b) Non-Trivial FDs
c) Axioms of sets
d) Armstrong axioms

VII. Consider the following statements:

S1: A relation in which every key contains only one attribute is in 2NF.

S2: In a 3NF relation, a non-key attribute may be transitively dependent on the primary key.

Choose the correct option:

a) S1: True; S2: True
b) S1: False; S2: True
c) S1: False; S2: False
d) S1: True; S2: False

VIII. Consider a B+ tree in which the maximum number of keys in an internal node is 5. What is the minimum number of keys in any non-root internal node?

a) 1
b) 2
c) 3
d) 4

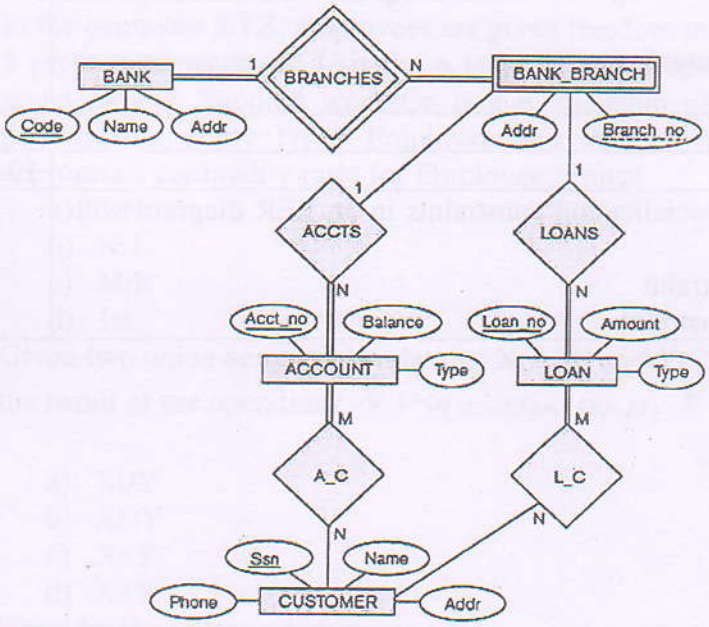
IX. Consider the following schedule S.

S: R1 (X); R2 (Z); R1 (Z); R3 (X); R3 (Y); W1 (X); W3 (Y); R2 (Y); W2 (Z);

In the precedence graph of S, the number of edges (both incoming and outgoing) of nodes T1 and T2, respectively are

a) 1, 1
b) 1, 2
c) 2, 1
d) 2, 2

X. If all transactions follow Two Phase Locking protocol, the resulting schedules will always be conflict serializable

	a) True b) False	
Q.2	Attempt any **two**	**10**
i)	a) Can identifying relationship of a weak entity be of degree greater than two? Give example to illustrate your answer b) Discuss entity integrity and referential integrity constraints with an example	5+5
ii)	a) Describe the three-schema architecture of a database system. Why are mappings between the schema levels necessary b) Discuss the advantages of database approach	5+5
iii)	a) Consider the following relational Schema: member(memb no, name) book(isbn, title, authors, publisher) borrowed(memb no, isbn, date) I. Write SQL DDL statement to create the member table II. Write SQL DML query to update the author of Database system book to "Korth" III. Write SQL DML query to find the member number and name of each member who has borrowed at least one book published by "McGraw-Hill". b) Convert the following ER diagram of the Bank database into a relational schema  	5+5
Q.3	Attempt any **two**	**10**
i)	a) Draw a state diagram and discuss the typical states that a transaction goes through during execution. b) Discuss the timestamp-based concurrency control protocol.	5+5
ii)	a) Given a relation R(A, B, C, D) and Functional Dependency set FD = { AB → CD, B → C }, determine whether the given R is in	5+5

	2NF? If not convert it into 2 NF. b) Construct the B+ Tree for the given input: A PARTS file with Part# as the key field includes records with the following Part# values:20, 11, 14, 25, 30. Suppose that the search field values are inserted in the give order in a B+ tree with order of internal node m=3 and order of leaf node m_{leaf}=2. Show how the tree will expand and what the final tree will look like.	
iii)	a) What is a view in SQL? Explain how views are created and used. Discuss any two advantages of using views in a database system with suitable examples b) What is a correlated subquery? Discuss with an example	5+5
Q.4	Attempt any **two**	**10**
i)	a) State the informal guidelines for relation schema design. Illustrate how violation of these guidelines may be harmful b) Discuss any 2 algorithms used for implementing SELECT relational algebra operation and circumstances under which each algorithm can be used	5+5
ii)	a) What is meant by conflict serializable schedule. Explain the concept of conflict serializability with an example. b) Discuss the recovery mechanism based on deferred update	5+5
iii)	a) What are ternary relationships? Can a ternary relationship be converted into a binary relationship? Justify with an example b) Explain the following Relational algebra Operations with examples: I. Left-outer join II. Union	
Q.5	Attempt the following	**10 (5+5)**
	Discuss the following specialization constraints in an EER diagram with suitable examples: a) Disjointness constraint b) Completeness constraint	